

LISTA DE EXERCÍCIOS – INTEGRAL IMPRÓPRIA

Turma: 2º Período Engenharia

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QUESTÃO 01

Determine se cada integral é convergente ou divergente. Calcule aquelas que são convergentes.

1. $\int_2^{\infty} \frac{1}{(x+3)^{3/2}} dx$

2. $\int_0^{\infty} e^{-x} dx$

3. $\int_{-\infty}^{\infty} x^3 dx$

4. $\int_{-\infty}^{\infty} \frac{1}{\sqrt[3]{w-5}} dw$

5. $\int_2^{\infty} \frac{1}{\sqrt{x+3}} dx$

6. $\int_{-\infty}^1 \frac{1}{(2x-3)^2} dx$

7. $\int_{\infty}^{-1} \frac{1}{\sqrt[3]{x-1}} dx$

8. $\int_{-\infty}^{\infty} x dx$

9. $\int_{-\infty}^{\infty} (2x^2 - x + 3) dx$

10. $\int_{-\infty}^0 e^{3x} dx$

11. $\int_1^{\infty} \sin \pi x dx$

12. $\int_0^{\infty} \frac{5}{2x+3} dx$

Questão 1. Determinar se é convergente ou divergente e calcular as convergentes.

Entrega: 31.07

Lembrete:

convergente = $1/x^c \rightarrow \exists \text{ Lim}$

divergente = $1/x \rightarrow \nexists \text{ Lim}$

Passo a passo p/ resolução:

$$1. \int_2^{\infty} \frac{1}{(x+3)^{3/2}} dx = \frac{1 dx}{\sqrt{(x+3)^3}} \rightarrow \lim_{b \rightarrow \infty} \int_2^b \frac{1}{(x+3)^{3/2}} dx \rightarrow 1. \frac{2}{3\sqrt{x+3}} \quad \frac{2}{3\sqrt{5}} - \frac{2}{3\sqrt{b+3}} \xrightarrow{b \rightarrow \infty} \frac{2}{3\sqrt{5}} - 0 = \frac{2\sqrt{5}}{15}$$

Relembrando: $dx \sqrt{x} = \frac{1}{2\sqrt{x}}$

$\frac{d}{dx} (x+3)^{3/2}$ R. Cadeia $u = x+3$ $\frac{3-1}{2} = \frac{1}{2}$
 $\frac{du}{dx} = 1$

$f(g(x)) = f'(g(x)) \cdot g'(x)$

$\frac{3}{2}(u)^{1/2} \cdot u' = \frac{3}{2}(x+3)^{1/2} \cdot 1$ R = $\frac{3\sqrt{x+3}}{2}$

$\frac{2}{3\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}}{15}$

Convergente

2. $\int_0^{\infty} e^{-x} dx \Rightarrow \lim_{b \rightarrow \infty} \int_0^b e^{-x} dx \rightarrow \frac{F(b) - F(0)}{e^x} = \frac{1}{e^0} - \frac{1}{e^b} \xrightarrow{b \rightarrow \infty} 1 - 0 = 1$ $e^0 = 1$ $\frac{1}{4} = 1$ $R = 1$ Convergente

3. $\int_{-\infty}^{\infty} x^3 dx = \int_{-\infty}^a x^3 dx + \int_a^{\infty} x^3 dx$

$\lim_{a \rightarrow \infty} \int_{-\infty}^a x^3 dx = \frac{x^4}{4} \Big|_{-\infty}^a = \frac{a^4}{4} - \frac{(-\infty)^4}{4} = \infty$

Divergente

4. $\int_{-\infty}^{\infty} \frac{1}{\sqrt[3]{w-5}} dw \rightarrow 1. \frac{3}{3\sqrt[3]{w-5}^2} \quad \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx$

$\frac{d}{dw} \sqrt[3]{w-5}$

$\frac{1-1}{3} \cdot \frac{3\sqrt[3]{w-5} + 1}{3} = -\frac{2}{3}$

$\lim_{a \rightarrow \infty} \int_{-\infty}^a \frac{1}{\sqrt[3]{w-5}} = \lim_{a \rightarrow \infty} \frac{3\sqrt[3]{w-5}^2}{3\sqrt[3]{w-5}^3} = \infty$

$\frac{1}{3} u^{-2/3} \cdot 1 = \frac{1}{3} (w-5)^{-2/3} = \frac{1}{3\sqrt[3]{(w-5)^2}}$

Divergente

5. $\int_2^{\infty} \frac{1}{\sqrt{x+3}} dx \Rightarrow 1. \frac{2}{2\sqrt{x+3}} \Rightarrow \lim_{a \rightarrow \infty} \int_2^a \frac{1}{\sqrt{x+3}} dx = \lim_{a \rightarrow \infty} \frac{F(b) - F(a)}{2\sqrt{x+3}}$

$\frac{d}{dx} \sqrt{x+3} = (x+3)^{-1/2} = \frac{1}{2\sqrt{x+3}}$

$\frac{1-1}{2} = -\frac{1}{2}$

$\frac{2\sqrt{5}}{2\sqrt{5}-\infty} = \infty$

Divergente

$$6. \int_{-\infty}^1 \frac{1}{(2x-3)^2} dx \Rightarrow \lim_{a \rightarrow -\infty} \int_a^1 \frac{1}{(2x-3)^2} dx \Rightarrow$$

Convergente

$$(2x-3)^2 dx$$

$$u=2$$

$$u=2x-3$$

$$du=2dx$$

$$\frac{du}{2}=dx$$

$$\int \frac{1}{2} \cdot \frac{1}{u^2} du$$

$$u^{-2+1} = \frac{u^{-1}}{-1} = -\frac{1}{u} + C$$

$$\frac{1}{2} \int \frac{1}{u^2} du = \frac{1}{2} \left[-\frac{1}{u} \right]_a^1$$

$$\begin{cases} -\infty = -\infty \\ 1 = -1 \\ x=1, u=-1 \end{cases}$$

$$\lim_{a \rightarrow -\infty} \frac{1}{2} \left[-\frac{1}{u} \right]_{-\infty}^{-1}$$

$$F(b) - F(a) = -\frac{1}{-1} - \left(-\frac{1}{-\infty} \right) = 1 - 0 = 1$$

$$7. \int_{-\infty}^{-1} \frac{1}{\sqrt[3]{x-1}} dx \Rightarrow \lim_{a \rightarrow -\infty} \int_a^{-1} \frac{1}{\sqrt[3]{x-1}} dx = \lim_{t \leq a} \text{intervalos errados}$$

Divergente

$$\sqrt[3]{x-1} = (x-1)^{\frac{1}{3}}$$

$$\frac{1}{3} u^{-\frac{2}{3}} \cdot 1 = \frac{1}{3 \sqrt[3]{x-1}^2}$$

$$8. \int_{-\infty}^{\infty} x dx \Rightarrow \int_a^{\infty} x dx + \int_{-\infty}^a x dx \Rightarrow \frac{x^2}{2} \Big|_a^{\infty} + \frac{x^2}{2} \Big|_{-\infty}^a = \infty \text{ Divergente}$$

$$9. \int_{-\infty}^{\infty} (2x^2 - x + 3) dx \Rightarrow \int_a^{\infty} (2x^2 - x + 3) dx + \int_{-\infty}^a (2x^2 - x + 3) dx$$

$$\left(\frac{2x^3}{3} - \frac{x^2}{2} + 3x \right) \Big|_a^{\infty} + \left(\frac{2x^3}{3} - \frac{x^2}{2} + 3x \right) \Big|_{-\infty}^a = \infty \text{ Divergente}$$

$$10. \int_{-\infty}^0 e^{3x} dx \Rightarrow \lim_{a \rightarrow -\infty} \int_a^0 e^{3x} dx \Rightarrow \lim_{a \rightarrow -\infty} \frac{e^{3x}}{3} \Big|_a^0 = \frac{1}{3} - 0 = \frac{1}{3} \text{ Convergente}$$

$$\frac{e^{kx}}{k}$$

$$11. \int_1^{\infty} \sin \pi x dx \Rightarrow \lim_{a \rightarrow \infty} \int_1^a \sin \pi x dx = \lim_{a \rightarrow \infty} \left[-\frac{1}{\pi} \cos \pi x \right]_1^a = -\frac{1}{\pi} \cos \pi a + \frac{1}{\pi} \cos \pi = \text{Diverge}$$

$$\int \sin \pi x dx = -\frac{1}{\pi} \cos \pi x + C$$

Periódica infinita
nunca atinge um valor finito

$$\frac{-1 \leq \cos \leq 1}{\pi} \rightarrow \lim \#$$

$$12. \int_0^{\infty} \frac{5}{2x+3} dx \Rightarrow \lim_{a \rightarrow \infty} \int_0^a \frac{5}{2x+3} dx = \lim_{a \rightarrow \infty} 5 \int_0^a \frac{1}{2x+3} dx = 5 \left[\frac{1}{2} \ln |2x+3| \right]_0^a = \frac{5}{2} \ln |2a+3| - \frac{5}{2} \ln |2 \cdot 0 + 3| = \infty \text{ Divergente}$$

$$\frac{1}{x} = \ln |x|$$

$$2x+3 = u$$

$$2dx = du$$

$$\frac{1}{2} du = dx$$

$$\frac{1}{2} du = dx$$

$$\frac{1}{2} du = dx$$

$$\nexists \lim$$

$$\frac{F(b) - F(a)}{2} = \frac{5 \ln |2a+3|}{2}$$

$$\frac{5 \ln \infty}{2} - \frac{5 \ln |2 \cdot 0 + 3|}{2} = \infty - 2.07 \dots = \infty$$